How Many Training Notes? Sample Efficiency of Logistic Regression, k-Nearest Neighbours and Random Forests for Forged-Banknote Detection

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Abstract

On the UCI Banknote Authentication benchmark, a tuned logistic regression already misclassifies only 0.56% $\pm$ 0.09% of notes in 5-fold cross-validation, so comparing methods by final accuracy alone would produce ties. We therefore ask how much labelled data each method needs: we compare logistic regression, k-nearest neighbours and random forests along learning curves from 20 to 800 training notes, using 50 random draws per size and tuning every method by cross-validation inside each draw. Early results show a crossover. Logistic regression is the most sample-efficient below about 100 notes (2.5% vs. 4.2% error for kNN at 50 notes), but it levels off near 0.6%, while kNN keeps improving to 0.13% at 800 notes. *[Final numbers, paired differences and the one-time test-set result to be added.]*

Feasibility summary for the early draft

- **Data acquired and loaded:** 1,372 $\times$ 5 raw table; 1,348 rows after removing 24 exact duplicates; 4 inputs, 1 binary output (Table 1).
- **Baseline result:** logistic regression (Module 3) tuned by stratified 5-fold CV (Module 4): **0.56% $\pm$ 0.09%** CV error; nested CV 0.74% $\pm$ 0.32% (Table 2). The baseline uses only material taught before 9 November.
- **Already running:** learning curves for all three methods (Figure 1, Table 3). The random forest (Module 7) is the planned extension; no headline result depends on it.
- **Untouched:** the 270-row stratified test set, reserved for one final evaluation.

1 Introduction *[outline]*

- Motivation: counterfeit detection is a cheap, low-dimensional classification task. On easy benchmarks, "which model is most accurate?" often ends in a tie, while "which model needs the fewest labelled examples?" still separates methods.
- Question: how many training notes does each method need to reach a given error rate (1%, 2%)? Is the answer different for a linear method (logistic regression), a local nonparametric one (kNN) and an ensemble (random forest)?
- Contributions: (i) a leakage-aware evaluation protocol (duplicate removal, scaling inside folds, an untouched test set, a check for several images per specimen); (ii) paired learning curves with error bars over 50 draws; (iii) an honest statement of where differences are ties.

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2 Related work *[outline]*

- Dataset origin: wavelet features of images of genuine and forged banknote-like specimens [1, 2].
- Learning curves for comparing classifiers: logistic regression tends to win on small samples and tree induction on large ones, and the curves can cross [3]; discriminative vs. generative sample efficiency [4]; a review of learning-curve shapes [5].
- Methods: logistic regression and cross-validation [6], nearest neighbours [7], random forests [8]; implementation [9].

3 Data

UCI Banknote Authentication [1] (CC BY 4.0) contains four features from a wavelet transform of each specimen image: variance, skewness, kurtosis and entropy. There is also a binary class. There are no missing values. We remove 24 exact duplicate rows, all from class 0. UCI does not document which class is forged, and secondary sources disagree. We treat class 1 as forged. Error and AUC do not depend on this choice; forged-class precision and recall would swap.

Table 1: Data size and shape at each stage. "Forged" is the share of class 1.

Stage	Rows	Inputs	Forged
Raw UCI file	1372	4	44.5%
After removing exact duplicates	1348	4	45.3%
Development set (80%)	1078	4	45.3%
Test set (20%, held out)	270	4	45.2%

First rows of the cleaned table (two per class):

Variance	Skewness	Kurtosis	Entropy	Class
3.622	8.666	-2.807	-0.447	0
4.546	8.167	-2.459	-1.462	0
-1.397	3.319	-1.393	-1.995	1
0.390	-0.143	-0.032	0.351	1

4 Method

Split. A stratified 80/20 split (seed 5420) gives 1,078 development rows and 270 test rows. The split is stored in `data/split.csv`, and the analysis code only ever loads the development rows.

Models and tuning. Each model is a scikit-learn pipeline, so standardisation is fitted on each training fold only. Logistic regression tunes $C \in \{10^{-3}, 10^{-2.5}, \ldots, 10^{3}\}$, kNN tunes $k \in \{1, 3, 5, 7, 9, 11, 15, 21, 31\}$, and the random forest (200 trees) tunes `max_features` $\in \{1, 2, 4\}$ and `min_samples_leaf` $\in \{1, 3\}$. Hyperparameters are always chosen by stratified 5-fold CV on training data only.

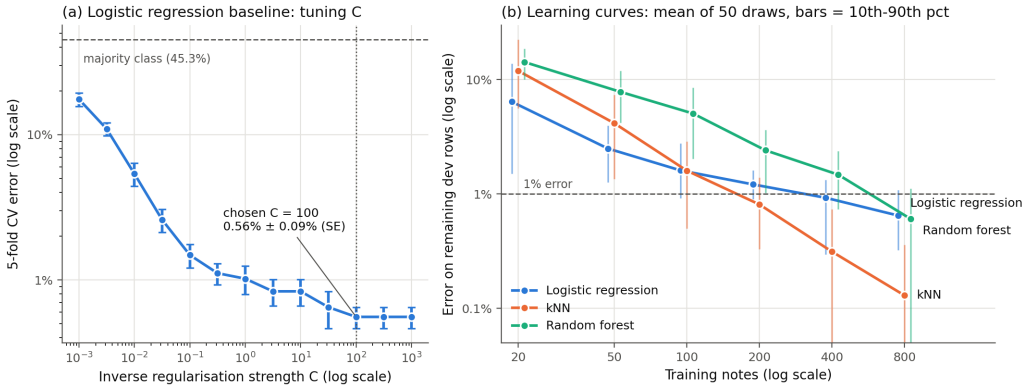
Learning curves. For $n \in \{20, 50, 100, 200, 400, 800\}$ we draw 50 stratified training sets of size n from the development data. In each draw we tune every method by 5-fold CV inside the training set and score it on all remaining development rows. All methods share the same draws, so the comparison is paired.

Specimen leakage check. The file has no specimen identifiers. However, class-1 rows show strong serial correlation in file order (lag-1 autocorrelation of skewness 0.62, against about 0 for class 0), which suggests several images per specimen. As a check, we re-ran CV with each run of 8 consecutive rows kept in one fold (`StratifiedGroupKFold`). The errors were unchanged: logistic regression 0.65% in both cases, kNN 0.09% in both cases.

5 Results so far

Table 2: Baseline on the development data (test set untouched). SE is the standard error of the five fold errors, SD/$\sqrt{5}$. The 95% interval is a Wilson interval on the pooled out-of-fold errors.

Model / estimate	Error	SE	Notes
Majority class	45.3%	–	–
Logistic regression, $C = 100$ (5-fold CV)	0.56%	0.09%	6/1078; 95% CI 0.26%–1.21%
Logistic regression, nested 5×5 CV	0.74%	0.32%	AUC 0.9999
Logistic regression, 10 repeats of 5-fold	0.54%	–	range 0.37%–0.65%



UCI Banknote Authentication (Lohweg 2012, CC BY 4.0); 1,078 development rows after de-duplication; test set (270 rows) unused.

Figure 1: (a) Logistic-regression CV error across the C grid, with ±1 SE bars. (b) Learning curves: mean error on held-out development rows over 50 draws; bars show the 10th–90th percentiles across draws.

Table 3: Learning curves: mean error (%) ± SE over 50 draws on the remaining development rows. Bold marks the lowest mean; gaps within about 2 SE (e.g. $n = 100$) are ties. †Module 7 extension.

n	Logistic regression	kNN	Random forest†
20	**6.39 ± 0.85**	11.89 ± 1.04	14.22 ± 0.71
50	**2.50 ± 0.20**	4.15 ± 0.35	7.81 ± 0.45
100	1.61 ± 0.11	**1.60 ± 0.15**	5.05 ± 0.34
200	1.21 ± 0.06	**0.81 ± 0.07**	2.41 ± 0.15
400	0.93 ± 0.05	**0.31 ± 0.04**	1.47 ± 0.09
800	0.65 ± 0.06	**0.13 ± 0.03**	0.60 ± 0.06

Reading so far. Logistic regression needs about 400 notes to reach 1% mean error, kNN about 200, and the random forest between 400 and 800. Cross-validation picks $k = 1$ at every size, which is consistent with near-duplicate images of the same specimen. Even so, the grouped check above shows no change in error.

6 Remaining sections *[outline]*

Experiments to finish (by 30 November).

- Add sizes $n \in \{10, 30, 150, 300, 600\}$; report paired differences (kNN − LR, RF − LR) with SE and the share of draws each method wins.
- Estimate n needed to reach 1% and 2% error for each method, with a bootstrap interval over draws.

- Extension (Module 7): tune the forest more fully (trees, OOB error), and add boosting if time allows.
- One final evaluation on the 270-row test set, with confusion matrices and ROC curves. Report it as a tie where the SE (≈0.4–0.6%) says so.

Discussion and limitations. Why a linear boundary saturates and k=1 keeps improving on these features. Conclusions are limited to this benchmark, not to real-world currency authentication. We cover the label-coding ambiguity, specimen leakage that cannot be fully tested, overlap between the evaluation sets of different draws, and the small test set.

Conclusion. One paragraph answering the sample-size question.

Contributions and AI use *[draft]*

Maanas built the data pipeline, the baseline and the learning-curve experiments. Advit will lead the random-forest extension and the evaluation plots. Both write the paper. We used an LLM assistant (Claude) for code generation, debugging and copy-editing. All experimental results were produced and checked by the authors. Code: https://github.com/maanasgoel/ese5420-banknote.

References

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